

Week 3: Confounding & Identification

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Learning Objectives

This week, we tackle the central challenge in observational causal inference: **confounding**. Our goal is to move from observing an association to making a credible causal claim. We will learn a rigorous, graphical method—the **backdoor criterion**—to identify the variables we must adjust for to achieve this. By the end of this module, you will be able to define confounding, use Directed Acyclic Graphs (DAGs) to find an appropriate adjustment set, and implement this logic in R.

Conceptual Overview

What is Confounding?

A **confounder** is a variable that is a common cause of both the treatment (A_i) and the outcome (Y_i). This shared cause creates a non-causal, “spurious” association between the treatment and outcome that can be mistaken for a true causal effect.

Exchangeability: The Key to Causal Inference

To estimate a causal effect, we need the treated and untreated groups to be **exchangeable**.

- **Unconditional Exchangeability:** This is the ideal state, achieved in a Randomized Controlled Trial (RCT). Randomization ensures that treatment assignment (A_i) is independent of the potential outcomes (Y_i^a). This means the treated and untreated groups are, on average, identical on all factors (measured and unmeasured) that might influence the outcome. We write this as:

$$Y_i^a \perp A_i \tag{1}$$

- **Conditional Exchangeability:** In observational studies, we cannot randomize. Instead, we rely on the assumption that we have measured all common causes of treatment and outcome (the confounders, X_i). If we adjust for this set of measured confounders, then within each level (or stratum) of X_i , the treatment assignment is “as good as random.” This crucial assumption is formally stated as:

$$Y_i^a \perp A_i \mid X_i \quad (2)$$

As we see in Equation 1 and Equation 2, the challenge is to identify the correct set of variables X_i to measure and adjust for.

Methodological Core

The Backdoor Criterion

The **backdoor criterion** is a graphical rule that uses a DAG to tell us which set of variables satisfies the conditional exchangeability assumption.

A set of variables X_i satisfies the **backdoor criterion** relative to a treatment A_i and an outcome Y_i if:

1. It blocks every backdoor path from A_i to Y_i .
2. It does not contain any variable that is a descendant of A_i .

A **backdoor path** is any path connecting A_i and Y_i that starts with an arrow pointing *into* A_i . To block a backdoor path, we must “condition on” a variable on that path.

Numerical Examples

Let’s illustrate with a scenario. We are studying the effect of a new medication (A_i) on heart disease (Y_i). We believe that age (X_i) is a confounder, as older patients are more likely to receive the medication and also more likely to have heart disease.

The DAG is: $A_i \leftarrow X_i \rightarrow Y_i$ and $A_i \rightarrow Y_i$.

- **Backdoor Path:** $A_i \leftarrow X_i \rightarrow Y_i$.
- **Adjustment Set:** To block this path, we must adjust for X_i .

Standardization by Hand

Suppose we have the data in Table 1.

Table 1

Group	Treatment ($A_i = 1$)
Young ($X_i = 0$)	10/40 (25%)
Old ($X_i = 1$)	40/60 (66.7%)

1. Crude (Biased) Association:

- Risk among treated: $(10 + 40)/(40 + 60) = 50/100 = 50\%$.
- Risk among untreated: $(5 + 20)/(20 + 80) = 25/100 = 25\%$.
- Crude Risk Difference: $50\% - 25\% = 25\%$. This is confounded.

2. Adjusted Estimate (Standardization):

We calculate the ATE by weighting the stratum-specific effects by the prevalence of each stratum.

- Proportion Young: $(40 + 20)/200 = 0.3$.
- Proportion Old: $(60 + 80)/200 = 0.7$.
- $ATE = \sum_x (E[Y_i|A_i = 1, X_i = x] - E[Y_i|A_i = 0, X_i = x])P(X_i = x)$
- $ATE = (0.25 - 0.25) \times 0.3 + (0.667 - 0.25) \times 0.7 \approx 29.2\%$.

The adjusted causal effect is a 29.2% increase in mortality, which differs from the crude estimate of 25%.

R Applications

We can use a regression model to perform the adjustment. The key is to include the confounder X_i in the model.

```
pacman::p_load(dagitty)
dag <- dagitty(
  "dag {
    A [exposure];
    Y [outcome];
    X;
    X -> A;
    X -> Y;
    A -> Y;
  }
")
plot(dag)
```

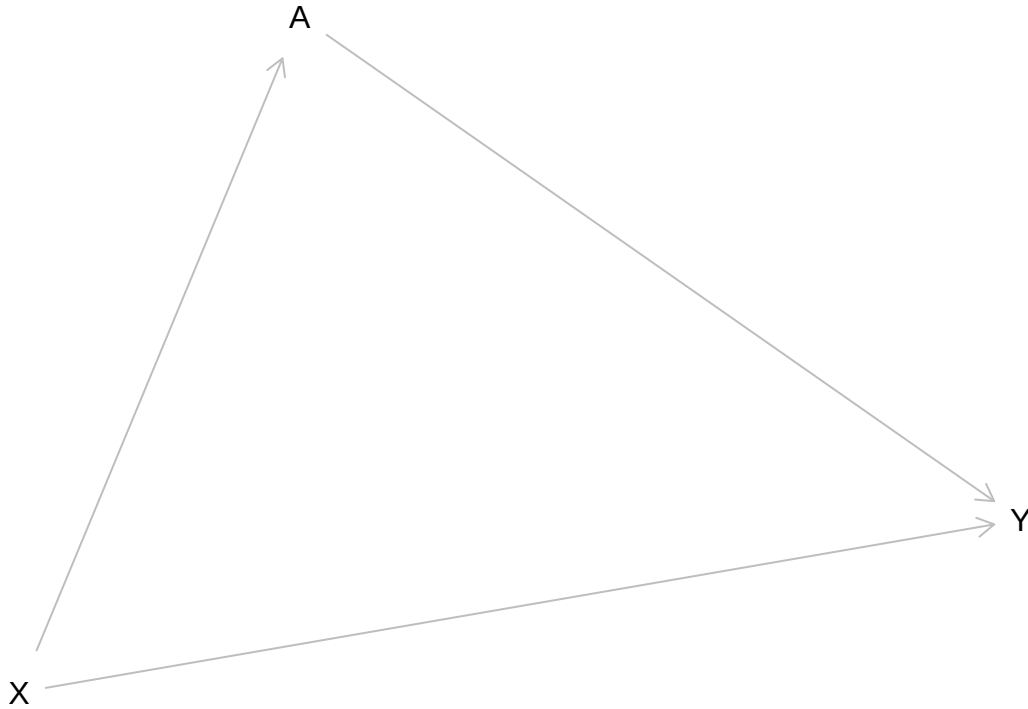


Figure 1: DAG showing confounding of the A-Y relationship by X

The DAG in Figure 1 shows that we must adjust for X.

Common Pitfalls

Adjusting for the wrong variable can create bias. These are “bad controls.”

- **Mediators:** A mediator (M_i) is on the causal pathway from A_i to Y_i ($A_i \rightarrow M_i \rightarrow Y_i$). Adjusting for a mediator blocks part of the causal effect you want to measure.
- **Colliders:** A collider is a variable caused by two other variables (e.g., $A_i \rightarrow Z_i \leftarrow Y_i$). Conditioning on a collider *opens* a non-causal path and induces bias.

Exercises

1. In an observational study, if we are certain that all confounders are measured, what core assumption from the potential outcomes framework (Equation 2) are we trying to satisfy?
2. Draw a DAG where X_i is a confounder and M_i is a mediator. What is the minimal sufficient adjustment set to estimate the total effect of A_i on Y_i ?

3. A study on exercise (A_i) and blood pressure (Y_i) adjusts for diet (X_i) and a genetic marker (Z_i). The true DAG is $A_i \leftarrow X_i \rightarrow Y_i$ and $A_i \rightarrow Z_i \leftarrow Y_i$. Is adjusting for Z_i a good idea? Why or why not?

Hands-On R Lab

Let's implement the adjustment in R.

```
# Simulate data based on our DAG
set.seed(123)
n <- 200
X <- rbinom(n, 1, 0.7) # Confounder (0=young, 1=old)
A <- rbinom(n, 1, prob = ifelse(X == 0, 0.4, 0.6)) # Treatment depends on X
Y <- rbinom(n, 1, prob = plogis(-1 + 0.5 * A + 1.5 * X)) # Outcome
df <- data.frame(A, Y, X = as.factor(X))

# 1. Crude (confounded) model
crude_model <- glm(Y ~ A, data = df, family = binomial())
summary(crude_model)
```

Call:

```
glm(formula = Y ~ A, family = binomial(), data = df)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.09909	0.19925	-0.497	0.619
A	0.40447	0.28470	1.421	0.155

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 276.76 on 199 degrees of freedom
Residual deviance: 274.73 on 198 degrees of freedom
AIC: 278.73

Number of Fisher Scoring iterations: 4

```
cat("Crude OR:", exp(coef(crude_model)['A']), "\n")
```

Crude OR: 1.498512

```
# 2. Adjusted (unconfounded) model
adjusted_model <- glm(Y ~ A + X, data = df, family = binomial())
summary(adjusted_model)
```

Call:

```
glm(formula = Y ~ A + X, family = binomial(), data = df)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.1886	0.3306	-3.596	0.000324	**
A	0.0540	0.3113	0.173	0.862295	
X1	1.7093	0.3710	4.607	4.09e-06	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 276.76 on 199 degrees of freedom
Residual deviance: 250.52 on 197 degrees of freedom
AIC: 256.52

Number of Fisher Scoring iterations: 4

```
cat("Adjusted OR:", exp(coef(adjusted_model)['A']), "\n")
```

Adjusted OR: 1.055485